

Karthik Krishnagiri Pachiyappan

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BUSINESS ANALYST • DATA-DRIVEN PROCESS IMPROVEMENT • ENTERPRISE STRATEGY

SUMMARY

I frame business problems quantitatively, test them against transactional data and lead the resulting solution into production, backed by 3 years of consulting at Accenture and a master's in Engineering Management. Across 3 enterprise programs spanning 90+ countries and a \$30M improvement portfolio, my analysis drove the strategic and tactical recommendations that cut technology cost 27%, lifted measured process performance 13% and carried user adoption to a 17% gain after go-live.

EDUCATION

M.S., Engineering Management | University of Massachusetts Amherst GPA: 3.7/4.0 | May 2026
(Graduate Certificate, Engineering Project Management & Leadership)

B.E., Production Engineering | PSG College of Technology GPA: 8.3/10.0 | May 2021

EXPERIENCE

Business Analyst | Accenture Jul 2021 – Aug 2024

- Identified process improvement opportunities across order, fulfillment and customer operations for a global client's \$250M expansion in 90+ countries, converting analysis into recommendations that lifted measured process performance 13%.
- Framed problems, developed hypotheses and tested them against operational data before committing to a solution, scoping feasibility and leading implementation across 24 improvement projects inside a \$30M portfolio that cut technology cost 27%.
- Queried transactional and master data in SQL to size problems, validate hypotheses and reconcile discrepancies across a 3-system landscape, building the quantitative evidence base behind every recommendation put to leadership.
- Partnered with IT, sales, finance and operations colleagues across 3 concurrent programs, running requirement workshops and negotiating agreement between competing process owners on changes to service quality, throughput and cost.
- Established master data governance workflows and harmonized customer, product and pricing data across regions, resolving discrepancies at their source rather than in reporting and reducing operating cost 18% for the client.
- Built executive KPI dashboards in Power BI and Tableau as the post-launch monitoring plan, giving leadership a standing view of adoption and process performance and carrying user adoption to a 17% gain across the portfolio.

Digital Marketing Intern | Zoho Corporation Apr 2021 – Jul 2021

- Analyzed real-time engagement and web analytics data to isolate where users dropped out of the funnel, then designed A/B tests on page content and campaign targeting to establish which changes actually moved conversion.
- Ran keyword and audience research with cross-functional marketing and sales teams to define the ideal B2B customer profile, sharpening lead-generation targeting and lifting search rankings and organic traffic for Zoho's SaaS products.

Supply Chain Intern | Hydac Jan 2021 – Apr 2021

- Built a KPI dashboard measuring assembly-unit efficiency at 57%, then built a Structural Equation Model on the plant's operating data to isolate which supplier, manufacturing and distribution drivers actually moved that number.
- Modeled a demand-driven inventory scenario (DDMRP) against historical demand and lead-time data, then defined a phased implementation roadmap targeting up to 70% assembly-unit efficiency within the following three quarters.

KEY PROJECTS

- Process Mining & Bottleneck Analysis (Python):** Reconstructed a synthetic 18,181-event log across 1,800 orders into 16 process variants using pandas and networkx, isolating where a 24.3-day median cycle was lost. Ranked backorder release at 18.6% of orders and 4.9 days ahead of credit blocks and compliance holds, and showed the operational measure belonged on fulfillment lead time rather than total cycle time, since payment terms, not operations, dominate the total.
- Order & Exception Status Dashboard:** Built a web dashboard (JavaScript, OpenUI5) tracking each order end to end, flagging credit, availability and backorder exceptions on arrival and replacing a manual routine that required IT escalation.
- Market Entry Analysis, WarmStop (UMass):** Sized a 30,000+ commuter market for a heated transit bench concept, ran SWOT and PESTLE on the competitive landscape and built a five-year financial model showing a ~\$1.1M revenue path.

CORE COMPETENCIES

- Analysis & Modeling:** Hypothesis development and A/B testing, process mining, structural equation modeling, demand forecasting, scenario and five-year financial modeling, market sizing, root-cause analysis, cost-benefit analysis, KPI definition.
- Data & Tools:** SQL, Python (pandas, networkx, matplotlib), Excel, PowerPoint, Power BI, Tableau, JavaScript and OpenUI5, enterprise ERP and CRM platforms, data reconciliation and validation, GenAI tools with response validation.
- Business & Consulting:** Requirements gathering and workshops, business case development, process redesign and mapping, master data governance and data quality, executive reporting, stakeholder facilitation and influencing.
- Execution & Partnership:** Solution development and feasibility scoping, implementation leadership, UAT and validation, monitoring plans, change management and training, Agile and hybrid delivery, cross-functional partnership.

LEADERSHIP & RECOGNITION

- Recognized by client and Accenture leadership as a top performer for delivery consistency across 3 years.
- Promoted to Student Supervisor at UMass, running 15+ staff per shift across 7 stations serving 7,000+ customers a day.
- Led the student board as Chairperson for 500+ students, Department of Production Engineering, PSG College of Technology.